



SDG — Complete Exam-Ready Answers

1. Define SDG

SDG stands for **Sustainable Development Goals**. It is a universal set of **17 goals** and **169 targets** adopted by **193 UN member states** on **September 25, 2015**, as part of the landmark document "**Transforming Our World: The 2030 Agenda for Sustainable Development**."

The SDGs were created as a **successor to the Millennium Development Goals (MDGs)** (2000–2015) and are designed to be achieved by the year **2030**.

Key characteristics of SDG:

- Adopted by the **United Nations (UN)**
- Applies to **all countries** — developed, developing, and underdeveloped
- Contains **17 Goals** and **169 specific targets**
- Built on the motto "**Leave No One Behind**"
- Addresses the three dimensions of sustainable development: **Economic, Social, and Environmental**
- Organized under the **5 P's framework**: People, Planet, Prosperity, Peace, and Partnership

The SDGs recognize that ending poverty and other deprivations must go hand-in-hand with strategies that improve health, education, reduce inequality, and spur economic growth — all while tackling climate change and working to preserve our oceans and forests.

2. All 17 SDG Goals (Under 5 P's Framework)



P1 — PEOPLE

(Goals focused on human dignity, ending poverty & ensuring healthy lives)

#	Goal	Explanation
1	No Poverty	End poverty in all its forms everywhere by ensuring equal rights to resources
2	Zero Hunger	End hunger, achieve food security, improve nutrition and promote sustainable agriculture
3	Good Health & Well-being	Ensure healthy lives and promote well-being for all people of all ages
4	Quality Education	Ensure inclusive, equitable, quality education and promote lifelong learning opportunities for all

#	Goal	Explanation
5	Gender Equality	Achieve gender equality and empower all women and girls to participate equally in society

P2 — PLANET

(Goals focused on protecting the environment and natural resources)

#	Goal	Explanation
6	Clean Water & Sanitation	Ensure availability and sustainable management of clean water and sanitation for all
12	Responsible Consumption & Production	Ensure sustainable consumption and production patterns to reduce waste and pollution
13	Climate Action	Take urgent action to combat climate change and its devastating impacts
14	Life Below Water	Conserve and sustainably use the oceans, seas and marine resources
15	Life on Land	Protect, restore and promote sustainable use of terrestrial ecosystems and halt biodiversity loss

P3 — PROSPERITY

(Goals focused on economic growth and well-being)

#	Goal	Explanation
7	Affordable & Clean Energy	Ensure access to affordable, reliable, sustainable and modern energy for all
8	Decent Work & Economic Growth	Promote sustained, inclusive economic growth, full employment and decent work for all
9	Industry, Innovation & Infrastructure	Build resilient infrastructure, promote inclusive industrialization and foster innovation
10	Reduced Inequalities	Reduce inequality within and among countries by empowering all regardless of age, sex or disability
11	Sustainable Cities & Communities	Make cities and human settlements inclusive, safe, resilient and sustainable

P4 — PEACE

(Goals focused on peaceful, just and inclusive societies)

#	Goal	Explanation
16	Peace, Justice & Strong Institutions	Promote peaceful and inclusive societies, provide access to justice for all and build strong institutions

P5 — PARTNERSHIP

(Goal focused on global cooperation)

#	Goal	Explanation
17	Partnerships for the Goals	Strengthen the means of implementation and revitalize the global partnership for sustainable development

3. How Computer Engineers Are Connected to SDG

Computer engineers play a **crucial and cross-cutting role** in achieving the SDGs. In the age of the **4th Industrial Revolution**, technology and digital innovation are the backbone of sustainable development.

Direct Connections:

SDG 4 — Quality Education Computer engineers build **e-learning platforms, digital classrooms, educational apps, and online libraries** that make quality education accessible to everyone, including rural and underprivileged communities. Example: Online exam systems, virtual universities.

SDG 3 — Good Health & Well-being Through **telemedicine applications, health monitoring systems, AI-based diagnostics, hospital management software** and electronic health records, computer engineers bring healthcare to remote areas.

SDG 8 — Decent Work & Economic Growth The **IT and software industry** is one of the largest employers globally. Computer engineers create jobs through software companies, startups, and freelancing platforms. Bangladesh's growing IT sector is a direct example.

SDG 9 — Industry, Innovation & Infrastructure Computer engineers drive **innovation through AI, IoT, robotics, cloud computing, and automation**, building the digital infrastructure that supports all industries.

SDG 11 — Sustainable Cities & Communities Computer engineers design and implement **smart city solutions** — intelligent traffic systems, digital governance, smart energy grids, waste management systems, and citizen service apps.

SDG 13 — Climate Action Through **green computing, energy-efficient data centers, carbon footprint tracking software**, and climate modeling, computer engineers help combat climate change. Example: Using one server instead of 10 reduces massive carbon emissions.

SDG 16 — Peace, Justice & Strong Institutions Computer engineers build **e-governance systems, cybersecurity frameworks, digital legal systems, transparency platforms**, and anti-corruption tools that promote justice.

SDG 17 — Partnerships for the Goals Computer engineers build the **global internet infrastructure, communication platforms, and data-sharing systems** that enable international collaboration and partnerships.



Summary Table:

SDG Goal Computer Engineer's Role

Goal 3	Telemedicine, health apps
Goal 4	E-learning, digital education
Goal 8	IT jobs, freelancing
Goal 9	AI, IoT, innovation
Goal 11	Smart city systems
Goal 13	Green computing
Goal 16	E-governance, cybersecurity
Goal 17	Internet infrastructure

Key Point: ACM Code of Ethics guides computer engineers to work responsibly and ethically while using technology to achieve these goals.

4. SDG in Bangladesh Perspective

Bangladesh is one of the **most committed developing nations** in implementing the SDGs. The country has aligned its national development plans with the 2030 agenda.



Bangladesh's Progress on Key SDGs:

SDG 1 — No Poverty Bangladesh has made remarkable progress in poverty reduction. The poverty rate dropped from over 40% in 2000 to below 20% in recent years. Social safety net programs like **VGD (Vulnerable Group Development)** and **allowances via mobile banking** directly support this goal.

SDG 2 — Zero Hunger Bangladesh achieved **food self-sufficiency** in rice production. Government programs distribute food to vulnerable populations. Agricultural technology improvements have boosted crop production.

SDG 3 — Good Health & Well-being Bangladesh has significantly reduced **maternal mortality, child mortality**, and increased **life expectancy**. Community clinics across the country provide healthcare at grassroots level. **Telemedicine** services have expanded in rural areas under Digital Bangladesh.

SDG 4 — Quality Education The government has ensured **free primary and secondary education**, distributed free textbooks, and established **multimedia classrooms** in schools. **Sheikh Russell Digital Labs** and e-learning platforms support digital education.

SDG 5 — Gender Equality Bangladesh has achieved **near-parity in primary and secondary education** for boys and girls. Women empowerment programs by organizations like **BRAC** and **Grameen Bank** have made Bangladesh a global model.

SDG 8 — Decent Work & Economic Growth Bangladesh's **RMG (Ready-Made Garments)** sector employs millions, especially women. The **IT/ITES sector** is growing rapidly, with 650,000+ freelancers making Bangladesh a top freelancing country. **bKash and mobile financial services** support economic inclusion.

SDG 9 — Industry, Innovation & Infrastructure Bangladesh **Hi-Tech Parks** (Kaliakoir Bangabandhu Hi-Tech City), **Sheikh Hasina Software Technology Park**, and ICT infrastructure development directly address this goal. Bangladesh connected to **submarine cables (SEA-ME-WE 4 & 5)** improving internet infrastructure.

SDG 11 — Sustainable Cities **Digital Bangladesh** and **Smart Bangladesh 2041** vision directly align with this goal. **Union Digital Centers (4,550+)**, smart city planning, and urban development programs support sustainable cities.

SDG 13 — Climate Action Bangladesh is one of the most **climate-vulnerable** countries. The government has a **Bangladesh Climate Change Strategy and Action Plan (BCCSAP)**. Bangladesh advocates strongly in global climate forums (COP) for climate justice.

SDG 17 — Partnership Bangladesh actively participates in global partnerships. The **a2i (Aspire to Innovate)** programme and international collaborations with UN agencies demonstrate this commitment.

Key Alignment: Digital Bangladesh → SDG

Digital Bangladesh Initiative	SDG Goal
Union Digital Centers	SDG 4, 9, 16
Mobile Banking (bKash)	SDG 1, 8, 10
e-Passport, Smart NID	SDG 16
Hi-Tech Parks	SDG 8, 9
Telemedicine	SDG 3
Multimedia Classrooms	SDG 4
Smart Bangladesh 2041	SDG 11, 9, 17

5. **Short Notes**

a) **Smart City**

Definition: A Smart City is an urban area that uses **Information and Communication Technology (ICT), data analytics, and digital infrastructure** to enhance the quality of life of its citizens, improve sustainability, and manage resources efficiently.

Key Features of a Smart City:

- **Smart Governance** — digital, transparent, paperless administration
- **Smart Economy** — cashless transactions, digital commerce, innovation
- **Smart Mobility** — intelligent traffic management, digital transport
- **Smart Environment** — IoT-based waste management, energy efficiency
- **Smart People** — digitally literate citizens
- **Smart Living** — e-health, e-education, safe housing

Components:

- High-speed internet connectivity
- CCTV and surveillance systems
- Smart energy grids
- Digital public service delivery
- Real-time data monitoring systems

Examples:

- Singapore (global model)
- Dhaka (aspiring under Smart Bangladesh 2041)
- Barcelona, Amsterdam (Europe)

Connection to SDG: Smart City directly supports **SDG Goal 11** (Sustainable Cities and Communities).

b) Smart Citizen

Definition: A Smart Citizen is a **digitally literate, socially aware, and technologically empowered individual** who actively participates in city governance, uses digital services, and contributes to sustainable development.

Characteristics of a Smart Citizen:

- Digitally literate — can use internet, apps, e-services
- Participates in **e-governance and digital democracy**
- Uses digital platforms for education, health, finance
- Environmentally conscious and responsible
- Aware of their **rights and responsibilities**
- Contributes to community development through technology

How to Become a Smart Citizen:

- Digital education and ICT training
- Access to internet and smart devices
- Awareness of government digital services
- Active participation in civic platforms

In Bangladesh Context:

- Citizens using **bKash, Nagad** for financial transactions

- Using **National Web Portal** for government services
- Online land records, birth certificates, passport services

Connection to SDG: Smart Citizen supports **SDG 4** (Quality Education) and **SDG 16** (Peace, Justice & Strong Institutions).

c) Smart Society

Definition: A Smart Society is a **connected, inclusive, equitable, and knowledge-based society** where technology bridges social gaps, empowers all members, and enables collective decision-making for sustainable development.

Characteristics of a Smart Society:

- **Inclusive** — no one is left behind regardless of gender, age, or class
- **Connected** — high-speed internet and digital platforms link all members
- **Knowledge-based** — decisions driven by data and information
- **Participatory** — citizens actively engage in social and political processes
- **Sustainable** — balances economic growth with social and environmental concerns

Pillars of Smart Society:

- Digital infrastructure
- Social equity and inclusion
- Education and innovation
- Community participation
- Ethical use of technology

Difference from Traditional Society:

Traditional Society	Smart Society
Paper-based	Digital-based
Slow communication	Real-time communication
Limited access	Universal access
Reactive	Proactive and data-driven

Connection to SDG: Smart Society supports **SDG 10** (Reduced Inequalities) and **SDG 17** (Partnership).

d) Smart Government

Definition: A Smart Government is a **digitally-driven, data-informed, transparent, and citizen-centric administration** that uses technology to deliver efficient public services, ensure accountability, and promote good governance.

Key Features:

- **E-governance** — all government services available online
- **Paperless administration** — digital records and documentation

- **Transparency and accountability** — open data and anti-corruption tools
- **Data-driven decisions** — policy-making based on real-time data
- **Citizen participation** — digital platforms for public feedback

Examples of Smart Government Initiatives:

- Online tax filing systems
- Digital birth/death registration
- e-Tendering for government procurement
- National helplines (e.g., Bangladesh's **333 helpline**)
- e-Passport systems
- Smart NID cards

In Bangladesh:

- **a2i Programme** (Aspire to Innovate) — government innovation program
- **Bangladesh National Portal** — centralized government information
- **Union Digital Centers** — grassroots government service delivery
- **e-GP (Electronic Government Procurement)** system

Connection to SDG: Smart Government directly supports **SDG 16** (Peace, Justice & Strong Institutions) and **SDG 17** (Partnerships).

e) Globalization

Definition: Globalization is the **process of increased interconnectedness and interdependence among countries** through the free flow of goods, services, capital, people, technology, and information across national borders, making the world function as a single integrated unit.

Key Dimensions of Globalization:

Dimension	Explanation
Economic	Free trade, multinational companies, global markets
Cultural	Exchange of culture, language, food, fashion, music
Political	International organizations, global governance (UN, WTO)
Technological	Internet, social media, global communication
Social	Migration, international education, global civil society

Positive Effects:

- Economic growth and job creation
- Technology transfer to developing countries
- Cultural exchange and diversity
- Better access to goods and services
- Spread of democracy and human rights

Negative Effects:

- Cultural homogenization (loss of local culture)
- Exploitation of cheap labor in developing countries
- Increased inequality between rich and poor nations

- Environmental degradation
- Brain drain from developing countries

Globalization & SDG Connection:

- SDG 17 (Partnerships) is essentially the SDG goal for globalization
- SDG promotes **positive globalization** — inclusive, sustainable, fair
- Digital globalization supports SDG 4 (education), SDG 9 (innovation)

In Bangladesh Context:

- RMG export connects Bangladesh to global markets
- Remittances from migrant workers
- IT/Freelancing connects Bangladeshi professionals globally
- Mobile phone arrived in Bangladesh in **1990s** — an example of technological globalization

f) Quality Education (SDG Goal 4)

Definition: Quality Education means ensuring **inclusive, equitable, and quality education** and promoting **lifelong learning opportunities** for all people, regardless of gender, age, disability, race, or economic status.

SDG 4 Full Name: *"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all"*

Key Targets of SDG Goal 4:

- Free primary and secondary education for all
- Equal access for women and girls
- Affordable vocational and higher education
- Increase number of qualified teachers
- Build and upgrade education facilities
- Promote digital and ICT literacy

What Makes Education "Quality"?

Dimension	Meaning
Accessible	Available to everyone regardless of location or income
Inclusive	Includes girls, disabled, minorities
Relevant	Curriculum matches real-world needs
Effective	Students actually learn and develop skills
Equitable	Equal opportunities for all

Role of Computer Engineers in Quality Education:

- Building **e-learning platforms** (Coursera, Khan Academy type)
- Developing **educational apps** for mobile learning
- Creating **digital libraries** and open-access resources
- Setting up **virtual classrooms** and video conferencing tools
- Designing **AI-based personalized learning** systems
- Building **examination and assessment management systems**

Quality Education in Bangladesh:

- Free textbook distribution to all students
- **Multimedia Classrooms** in schools
- **Sheikh Russell Digital Labs**
- **Online university education**
- **LEDP (Learning and Earning Development Project)** for IT skills
- **National Curriculum and Textbook Board (NCTB)** digital content

Why Quality Education is Key to All SDGs:

- Educated people can fight poverty (SDG 1)
- Education promotes gender equality (SDG 5)
- Educated engineers can solve environmental problems (SDG 13)
- Digital education supports innovation (SDG 9)
- Education is the foundation of Smart Citizens, Smart Society

Ma'am's Note Connection: "SDG Goals (17) → যারা BD-র মানুষকে education-এ এগিয়ে নিতে চায়" — SDG's core mission is to bring every Bangladeshi forward through education and digital empowerment.



Quick Revision Summary

Topic	Key Line
SDG Definition	17 goals, 169 targets, UN 2015, deadline 2030
Motto	"Leave No One Behind"
5 P's	People, Planet, Prosperity, Peace, Partnership
Computer Engineers	Build digital tools for health, education, governance, climate
Bangladesh	Digital BD aligns with SDG 4, 8, 9, 11, 17
Smart City	ICT-driven urban management for quality of life
Smart Citizen	Digitally literate, participatory individual
Smart Society	Inclusive, connected, knowledge-based community
Smart Government	Transparent, digital, citizen-centric administration
Globalization	World as one unit — economic, cultural, tech integration
Quality Education	Inclusive, equitable, lifelong learning for all



Intellectual Property (IP) — Complete Exam Preparation

1. Definition of Intellectual Property

Intellectual Property (IP) refers to **creations of the human mind** — inventions, literary and artistic works, designs, symbols, names, and images used in commerce — that are protected by law, giving creators **exclusive rights** over their use and distribution.

The word "**Intellectual**" comes from "**Intellect**" meaning the human mind/brain. So IP literally means "**Property created by the mind/brain.**"

In your professor's words:

"IP → মেধা (talent/intellect) + Skillness" "Google-র logo চাইলেই use করতে পারবে না — this is Intellectual Property"

Formal Definition: Intellectual Property is a category of **intangible property** that includes creations of the intellect for which exclusive rights are recognized under law. IP rights allow creators to benefit from their own work and investment in creation.

Key Features of IP:

- It is **intangible** — you cannot touch it physically
 - It is a product of **human creativity and intellect**
 - Protected by **national and international laws**
 - Gives the creator **exclusive rights** to use, sell, or license
 - Has a **limited time period** of protection (except trademarks)
 - Can be **bought, sold, or transferred** like physical property
-

2. Historical Evolution of Property Concept (Professor's Key Point!)

Your professor specifically traced the **origin of property concept** — this is very important for your exam!

Stage 1 — **Hunting Society (Collecting Stage)**

- No concept of "**mine**" or "**yours**"

- Everything was **collective/shared** — "আমরা" (we) concept
- People collected fruits and hunted together
- **No individual property existed**
- Karl Marx called this "**Primitive Communism**"
- Property idea began here as a "**we**" concept

Stage 2 — **Fishing & Hunting Stage**

- Humans began using **tools** (bow and arrow — first creativity!)
- This was the **first human invention** — beginning of IP concept
- Stones were gathered to make weapons
- First signs of **individual skill and creativity**

Stage 3 — **Agriculture Society**

- Concept of **land ownership** emerged
- "**আমার জমি**" (my land) — property became more personal
- Still mostly community-based

Stage 4 — **Industrial Revolution (1760 onwards)**

- **Machines and inventions** became valuable
- Need to protect inventions → **Patent system** emerged
- Large-scale production → brand identity → **Trademark** emerged

Stage 5 — **Capitalistic Society**

- "**আমি**" (I/me) concept fully developed
- **Individual ownership** of ideas, creativity, inventions
- IP formally recognized and legally protected
- "**I, Me → from Capitalistic Society**" (as noted by professor)

Stage 6 — **4th Industrial Revolution (Current)**

- **Digital creativity** — software, algorithms, digital art
- **AI-generated content** — new IP challenges
- **Google logo, Facebook interface, software code** — all IP
- IP becomes more important than ever before
- **Computer Engineers must know IP** (ACM Code connection)

3. **Types of Intellectual Property (Main Categories)**

① Copyright ©

Definition: Protects **original creative works** such as books, music, films, software, paintings, articles, and photographs.

Key Points:

- Automatically granted upon creation — **no registration needed**
- Lasts for the **creator's lifetime + 50 to 70 years**
- Prevents unauthorized copying, distribution, or performance
- Example: A song you compose is automatically your copyright

Examples:

- A novel written by a Bangladeshi author
 - Source code written by a computer engineer
 - A photograph taken by a photographer
 - A film produced by a director
-

② Patent

Definition: Protects **inventions and innovations** — new products, processes, or technologies.

Key Points:

- Must be **registered** with a patent office
- Lasts **20 years** from the filing date
- Gives inventor **exclusive right** to make, use, sell the invention
- After 20 years → enters **public domain**
- Invention must be **novel, useful, and non-obvious**

Examples:

- A new medicine formula
 - A new machine design
 - A new manufacturing process
 - James Watt's Steam Engine (1760) — would be patented today
-

③ Trademark ™ / ®

Definition: Protects **brand identifiers** — names, logos, slogans, symbols, colors that identify a business.

Key Points:

- Can last **indefinitely** as long as it's actively used and renewed
- Prevents consumer confusion in the marketplace
- ™ = unregistered trademark; ® = registered trademark

Examples (Professor's Example!):

- **Google's logo** → cannot use without permission ✓
 - Nike's "Just Do It" slogan
 - Coca-Cola's distinctive lettering
 - Apple's bitten apple logo
 - Grameenphone's logo in Bangladesh
-

④ Trade Secret

Definition: Protects **confidential business information** that gives a competitive advantage.

Key Points:

- **No registration required** — protected as long as secrecy is maintained
- Can last **forever** if the secret is kept
- Employees must sign **Non-Disclosure Agreements (NDA)**

Examples:

- Coca-Cola's secret formula
 - Google's search algorithm
 - KFC's recipe (11 herbs and spices)
 - A software company's proprietary algorithm
-

⑤ Industrial Design

Definition: Protects the **visual/aesthetic aspects** of a product — its shape, color, pattern, or texture.

Key Points:

- Protects the **look** of a product, not its function
- Usually lasts **10–25 years**
- Must be new and original

Examples:

- The unique shape of iPhone
 - A car's body design
 - The shape of a Coca-Cola bottle
 - A unique furniture design
-

⑥ Geographical Indication (GI)

Definition: Protects products that have a **specific geographical origin** and possess qualities or reputation due to that origin.

Key Points:

- Ensures quality associated with a region is not misused

- Cannot be used by producers outside that region

Examples:

- Muslin from Dhaka (Bangladesh) — GI protected
 - Darjeeling Tea (India)
 - Champagne (France)
 - Basmati Rice (India/Pakistan)
-

4. IP and the 4th Industrial Revolution (Very Important!)

Your professor emphasized: "IP → connected to 4th Industrial Revolution (very important)"

What is the 4th Industrial Revolution?

Revolution	Year	Key Feature
1st	1760	Steam Engine (James Watt)
2nd	1860	Electricity, mass production
3rd	1970	Computer, digital technology
4th	2000+	AI, IoT, Robotics, Big Data

Why IP is Critical in the 4th Industrial Revolution:

① Software and Code

- Every software program is protected by **copyright**
- Operating systems, apps, games — all IP protected
- A computer engineer's code is their intellectual property

② AI and Algorithms

- AI algorithms can be **patented**
- Machine learning models represent significant IP value
- Who owns AI-generated content? — New IP challenge

③ Digital Branding

- Company logos, websites, UI designs — all **trademarked**
- Example: Google's logo, Facebook's interface, YouTube's play button

④ Data as IP

- In the digital age, **data is a new form of property**
- Databases are protected under copyright law
- Personal data privacy connects to IP concepts

⑤ Open Source vs IP

- Open source software challenges traditional IP concepts
- Linux, Python — freely available but still IP regulated

Computer Engineers and IP:

What Engineers Create IP Protection Type

Software code	Copyright
New algorithm/invention	Patent
Company logo/brand	Trademark
Business algorithm	Trade Secret
UI/UX design	Industrial Design

ACM Code Connection: Computer engineers must **respect IP rights** of others and protect their own creations ethically. This is part of the **ACM Code of Ethics** — a professional obligation.

5. International IP Framework

Organization/Agreement	Role
WIPO (World Intellectual Property Organization)	UN agency governing global IP
TRIPS Agreement	WTO's trade-related IP rights agreement
Berne Convention	International copyright protection
Paris Convention	Protection of patents and trademarks
DPDT (Bangladesh)	Department of Patents, Designs and Trademarks

6. Why IP Matters — Importance

Reason	Explanation
Encourages Innovation	Rewards creators for their effort and investment
Economic Growth	IP-intensive industries create jobs and GDP
Consumer Protection	Prevents counterfeit and fake products
Cultural Development	Supports artists, authors, musicians
Fair Competition	Prevents businesses from stealing each other's work
National Development	Countries with strong IP laws attract investment

7. Exam-Ready Short Note on IP

"Intellectual Property (IP) refers to creations of the human mind that are protected by law. The concept of property originated from the **Hunting Stage** of human society, where the idea of collective ownership (we/আমরা) first emerged. With the development of **Capitalistic Society**, the concept of individual ownership (I/আমি) became prominent. IP today covers copyright, patents,

trademarks, trade secrets, industrial designs, and geographical indications. In the era of the **4th Industrial Revolution**, IP has become extremely significant as digital creations like software, algorithms, AI, and digital brands (e.g., Google's logo) are all protected intellectual property. Computer engineers must be aware of IP laws to protect their own creations and respect others' rights, as guided by the **ACM Code of Ethics**. International bodies like **WIPO** and agreements like **TRIPS** govern IP protection globally."

8. IP vs Public Domain

Aspect	Intellectual Property	Public Domain
Protection	Legally protected	No protection
Use	Requires permission	Anyone can use freely
Duration	Limited time	After IP expires
Example	Current Disney films	Shakespeare's works

9. ⚡ Common IP Violations (White Collar Crime Connection!)

Your professor connected **White Collar Crime** with IP — very relevant!

- **Piracy** — copying movies, music, software without permission
- **Plagiarism** — using someone's written work without credit
- **Counterfeiting** — making fake branded products
- **Hacking** — stealing proprietary code or data
- **Patent infringement** — using a patented invention without license

Professor's Note: "White Collar Criminal → Hacker" — Digital IP theft is a modern form of white collar crime that is hard to catch and punish.



Quick Revision Table

Type	What it Protects	Duration	Registration?
Copyright	Creative works	Life + 50-70 years	No (automatic)
Patent	Inventions	20 years	Yes
Trademark	Brand identity	Indefinite (renewable)	Yes
Trade Secret	Confidential info	Indefinite (if secret)	No
Industrial Design	Product appearance	10-25 years	Yes
GI	Regional products	Indefinite	Yes



Key Points for Exam (Don't Forget!)

Point	Detail
Origin	Property concept → Hunting Stage → "we" concept
Individual ownership	Started in Capitalistic Society → "I/Me" concept
Professor's example	Google logo = Intellectual Property
IP =	স্বৈয়া + Skillness
4th IR connection	Software, AI, digital branding all = IP
International body	WIPO (UN agency)
Bangladesh body	DPDT (Department of Patents, Designs and Trademarks)
ACM Code link	Engineers must respect and protect IP ethically
White collar crime	IP theft = modern white collar crime
Most important types	Copyright, Patent, Trademark (learn all 6)

Digital Bangladesh — Complete Exam Preparation

1. Definition of Digital Bangladesh

Digital Bangladesh is a **transformative national vision** declared by the **Government of Bangladesh** in **2008** under the leadership of **Sheikh Hasina**, with a target year of **2021** — coinciding with Bangladesh's **50th Independence Anniversary**.

It aimed to harness **Information and Communication Technology (ICT)** to build a **technology-driven, knowledge-based, modern Bangladesh** where government services, education, healthcare, commerce, and daily life are all powered by digital technology.

Core Philosophy:

"Use technology to bring government services to the doorstep of every citizen, reduce corruption, ensure transparency, and accelerate development."

Key Fact: Digital Bangladesh later evolved into the **Smart Bangladesh Vision 2041**, taking the digital transformation to the next level.

2. Evolution of Technology in Bangladesh (Professor's Key Timeline!)

Your professor specifically discussed the **history of mobile phones in Bangladesh** — this is very exam relevant!

Era 1 — Land Phone Era (Before 1990s)

- Bangladesh depended entirely on **land/telephone lines**
 - Call rates were **distance-wise** — local calls cheaper, long-distance expensive
 - One land phone served **2-3 houses** in a family
 - Very limited connectivity — mostly urban, government offices
-

Era 2 — Mobile Phone Arrives (1990s)

- Mobile phones came to Bangladesh in the **early 1990s**
- Price was extremely high — **80,000 to 90,000 taka**

- **No SIM card** system — phones were directly connected
 - Only **highly executive/elite people** could afford them
 - Very limited network coverage
 - Still essentially a luxury item
-

Era 3 — Slow Adoption (2000–2005)

- By **2000**, prices started dropping slowly
 - **A few more people** began using mobile phones
 - SIM cards were introduced
 - Network operators like **Grameenphone** began expanding
 - Still not affordable for the general public
-

Era 4 — Mass Spread (2006–2009)

- From **2006**, mobile phones spread rapidly
 - **Repair shops, assembly centers** emerged across the country
 - Chinese low-cost phones entered the market
 - Mobile became **repairable and affordable**
 - Rural areas started getting network coverage
-

Era 5 — Universal Access (2010 onwards)

- By **2010**, **almost everyone had a mobile phone**
 - Smartphones began entering the market
 - **Internet on mobile** became accessible
 - Mobile banking (**bKash launched 2011**) revolutionized finance
 - Bangladesh connected to **submarine cable** improving internet speed
-

Era 6 — Digital Bangladesh (2008–2021)

- Government declared **Digital Bangladesh vision**
 - ICT became a national priority
 - **Union Digital Centers, e-governance, smart NID, e-passport** launched
 - Bangladesh became one of top **freelancing countries**
 - IT exports grew significantly
-

Era 7 — Smart Bangladesh (2041 Vision)

- Building on Digital Bangladesh
- AI, IoT, robotics integration
- Fully cashless economy
- Smart cities, smart governance

3. Four Pillars of Digital Bangladesh

(Professor noted: "Digital Bangladesh → 4ঐ point")

Pillar 1 — Human Resource Development

- Building a **skilled, ICT-literate workforce**
- ICT education from school level
- Training programs for freelancers and IT professionals
- **Sheikh Russell Digital Labs** in schools
- **LEDP (Learning and Earning Development Project)**
- University-level IT education expansion

Pillar 2 — People's Connectivity

- Internet and telecom access for **all citizens**
- Connecting rural areas through mobile networks
- **Union Digital Centers** in 4,550+ unions
- Submarine cable connectivity (SEA-ME-WE 4 & 5)
- Broadband internet across the country
- Affordable smartphones and internet packages

Pillar 3 — Digital Government (E-Governance)

- All government services available **online**
- **National Web Portal** — centralized information hub
- **e-Tendering** — government procurement online
- **Smart NID** — biometric national identity
- **e-Passport** — modern biometric passport
- Land records, birth certificates online
- **National Helpline 333** — digital citizen service

Pillar 4 — IT Industry Promotion

- Growing the **IT/ITES export sector**
 - **Bangladesh Hi-Tech Park Authority (BHTPA)**
 - **Kaliakoir Bangabandhu Hi-Tech City**
 - Special Economic Zones for tech companies
 - Startup ecosystem in Dhaka
 - IT export incentives and tax benefits
-

4. Role of Computer Engineers in Digital Bangladesh

(Professor specifically mentioned this — very important for exam!)

Computer engineers are the **backbone of Digital Bangladesh**. Every digital initiative requires skilled computer engineers to design, build, maintain, and improve the systems.

Specific Roles:

① Building E-Government Systems

- Designing government websites and portals
- Building **e-service platforms** for birth certificates, land records, passports
- Creating **e-tendering systems** for government procurement
- Developing **National ID management systems**

② Mobile Financial Services

- Computer engineers built **bKash, Nagad, Rocket** platforms
- These systems transformed financial inclusion in Bangladesh
- Over 50 million mobile banking users — all powered by software engineers

③ Education Technology

- Building **e-learning platforms** for schools and universities
- Developing **digital content** for multimedia classrooms
- Creating **online examination systems**
- Platforms for digital textbooks distribution

④ Healthcare Technology

- **Telemedicine applications** connecting doctors to rural patients
- Hospital management software systems
- Health monitoring apps
- Digital prescription systems

⑤ Smart City Infrastructure

- Designing **intelligent traffic management systems**
- Building **CCTV and surveillance networks**
- Creating **smart utility management** (electricity, water)
- Developing **city management dashboards**

⑥ Cybersecurity

- Protecting government digital infrastructure
- Securing mobile banking transactions
- Preventing digital fraud and cybercrime

- Ensuring data privacy of citizens

⑦ IT Industry and Freelancing

- Computer engineers **export IT services** globally
- Bangladesh has **650,000+ freelancers** — mostly computer engineers
- Software companies earning **foreign currency** for Bangladesh
- Creating jobs in the digital economy

⑧ Fulfilling SDG Goals through Digital BD

- SDG 4 (Education) → e-learning platforms
- SDG 3 (Health) → telemedicine
- SDG 8 (Work) → IT jobs and freelancing
- SDG 11 (Smart Cities) → smart city tech
- SDG 16 (Governance) → e-governance systems



Summary Table: Computer Engineers in Digital Bangladesh

Area	Contribution
E-Governance	Portals, NID, e-passport systems
Mobile Banking	bKash, Nagad, Rocket platforms
Education	E-learning, digital classrooms
Healthcare	Telemedicine, health apps
Smart City	Traffic, surveillance, utilities
Cybersecurity	Data protection, fraud prevention
IT Export	Freelancing, software companies
SDG Achievement	Tech solutions for all 17 goals

5. Short Notes

a) Smart City (in Digital Bangladesh Context)

Definition: A Smart City is an **ICT-powered urban area** where technology is used to manage resources, deliver services, and improve the quality of life of citizens efficiently and sustainably.

Smart City Features in Bangladesh Context:

- **Digital governance** — all city services online
- **Intelligent transportation** — smart traffic management
- **Smart utilities** — digital electricity and water management
- **Public safety** — CCTV networks, emergency response systems
- **Digital commerce** — cashless transactions everywhere
- **Smart waste management** — IoT-based collection systems

Smart Bangladesh 2041 Smart City Vision:

- Dhaka, Chittagong, Sylhet being developed as smart cities
- Union Digital Centers as grassroots smart service points
- Digital infrastructure investment across urban areas

Connection: Smart City = SDG Goal 11 + Digital Bangladesh Vision

b) Smart Citizen (in Digital Bangladesh Context)

Definition: A Smart Citizen in the context of Digital Bangladesh is a **digitally literate, ICT-empowered individual** who can access government services online, participate in digital economy, and contribute to national development through technology.

Characteristics:

- Can use internet, smartphones, and digital apps
- Uses **e-government services** — NID, passport, land records online
- Participates in **mobile banking** — bKash, Nagad
- Engages in **digital commerce** — online shopping, digital payments
- Has **digital awareness** — knows cybersecurity basics
- Can access **e-education and e-health** services

How Digital Bangladesh Creates Smart Citizens:

- ICT training programs across the country
 - Union Digital Centers assist citizens with digital services
 - Digital literacy programs in schools
 - Free internet access points in public spaces
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c) Smart Society (in Digital Bangladesh Context)

Definition: Smart Society in the context of Digital Bangladesh is a **connected, inclusive, technology-enabled community** where digital tools bridge social and economic gaps, and every citizen has equal access to opportunities.

Features:

- **Financially inclusive** — mobile banking reaches unbanked population
- **Educationally connected** — e-learning for all
- **Digitally aware** — cybersecurity and digital rights knowledge
- **Economically empowered** — freelancing, e-commerce opportunities
- **Socially equitable** — technology reduces urban-rural divide

Examples in Bangladesh:

- Rural women accessing government allowances through mobile banking
- Students in remote areas accessing online education
- Farmers getting market prices through mobile internet
- Garment workers receiving salaries through mobile financial services

d) Smart Government (in Digital Bangladesh Context)

Definition: Smart Government in Digital Bangladesh means a **transparent, efficient, digital-first administration** that delivers services to citizens online, reduces corruption through technology, and makes data-driven policy decisions.

Key Initiatives:

Initiative	Description
National Web Portal	All government information centralized online
e-Tendering (e-GP)	Government procurement fully online
Smart NID	Biometric national identity card
e-Passport	Modern biometric passport system
Union Digital Centers	Government services at grassroots level
333 Helpline	Digital citizen service hotline
a2i Programme	Government innovation and digitization program
e-Mutation	Land transfer records online
Birth Registration	Online birth and death registration

Benefits:

- Reduced corruption through digital transparency
- Faster service delivery — no need to visit offices
- Cost reduction for both government and citizens
- Better data for policy-making
- Accountability through digital records

e) Cashless Society (Professor's Exam Point!)

Definition: A **Cashless Society** is one where financial transactions are conducted through **digital means** — mobile banking, credit/debit cards, internet banking, and digital wallets — rather than physical cash.

Why Professor mentioned it for Computer Engineers:

"Cashless society — exam mostly for computer engineers" Because computer engineers **build and maintain** all the systems that make cashless transactions possible.

Bangladesh's Journey to Cashless Society:

Year	Development
1990s	Land phones, no digital payments
2000s	ATM cards introduced
2011	bKash launched — mobile money revolution
2015	Nagad, Rocket expanded mobile banking
2020	COVID accelerated digital payments
2021	Government payments via mobile — salaries, allowances

Year **Development**
2041 **Smart Bangladesh — fully cashless target**
Key Mobile Financial Services (MFS) in Bangladesh:

- **bKash** — most popular, 50M+ users
- **Nagad** — government-backed, growing rapidly
- **Rocket** (Dutch-Bangla Bank)
- **mCash** (Islami Bank)
- **SureCash**

Role of Computer Engineers in Cashless Society:

- Building **secure payment gateways**
- Developing **mobile banking apps**
- Creating **fraud detection systems**
- Ensuring **encryption and data security**
- Building **merchant payment systems**
- Developing **QR code payment platforms**

Benefits of Cashless Society:

- Reduces black money and corruption
- Faster and more convenient transactions
- Financial inclusion for unbanked population
- Better tax collection for government
- Reduces cost of printing and managing cash
- Supports Digital Bangladesh and Smart Bangladesh goals

Challenges:

- Digital literacy required
- Internet connectivity needed
- Cybersecurity risks
- Not everyone has smartphones

6. Digital Bangladesh → Smart Bangladesh 2041

Aspect	Digital Bangladesh (2021)	Smart Bangladesh (2041)
Focus	ICT infrastructure	AI, IoT, advanced tech
Economy	Digital economy	Smart economy
Government	E-governance	Fully automated governance
Education	Digital classrooms	AI-powered personalized learning
Healthcare	Telemedicine	AI diagnostics, smart hospitals
Cities	Digital cities	Fully smart cities
Payment	Mobile banking	Fully cashless

Aspect	Digital Bangladesh (2021)	Smart Bangladesh (2041)
Citizens	Digitally literate	Smart, tech-empowered

4 Pillars of Smart Bangladesh 2041:

1. **Smart Citizen** — tech-savvy, digitally empowered population
2. **Smart Economy** — cashless, innovation-driven, AI-powered
3. **Smart Government** — fully automated, paperless
4. **Smart Society** — connected, equitable, sustainable

7. Digital Bangladesh & SDG Connection

Digital BD Initiative	SDG Goal
e-Education platforms	SDG 4 — Quality Education
Telemedicine	SDG 3 — Good Health
Mobile banking	SDG 1 — No Poverty, SDG 10 — Reduced Inequality
IT jobs/freelancing	SDG 8 — Decent Work
Hi-Tech Parks	SDG 9 — Industry & Innovation
Union Digital Centers	SDG 11 — Sustainable Cities
e-Governance	SDG 16 — Peace & Justice
International IT export	SDG 17 — Partnership



Quick Revision Summary

Topic	Key Points
Digital Bangladesh	Vision 2008, target 2021, 4 pillars
4 Pillars	Human Resource, Connectivity, E-Governance, IT Industry
Mobile Timeline	1990s (80-90k taka) → 2000 (slow spread) → 2006 (mass) → 2010 (everyone)
Land Phone	Distance-wise call rates, one phone per 2-3 houses
Smart Bangladesh	2041 vision, 4 pillars: Citizen, Economy, Government, Society
Cashless Society	bKash, Nagad, digital payments — built by computer engineers
Computer Engineers	Build everything — e-gov, m-banking, smart city, cybersecurity
SDG Connection	Digital BD supports SDG 4, 8, 9, 11, 16, 17
Key Platforms	bKash (2011), Union Digital Centers (4550+), e-Passport, Smart NID
Global Rank	Bangladesh top freelancing country, 650,000+ freelancers



Professor's Exam Hint: Focus on — *Digital Bangladesh 4 points + Smart Technology + Smart City + Role of Computer Engineers + Cashless Society + Mobile phone evolution timeline in Bangladesh*